

Weekly Report Content

General Information

- Group ID: 53
- Group name: Echoes of the Ashen Vault.
- Project name: Echoes of the Ashen Vault (Dungeon Game).
- Date range: 29.06.2026 - 04.07.2026

Tasks completed this week

Since this is the first weekly report, we will include all tasks from the first 4 weeks.

25125027 - Nguyễn Tấn Minh:

Week 01 & 02

- Create [detailed game description](#) and [OOP architecture](#).
- Setup project, folder structure, and CMake and link dependencies.

Week 03

- Implement [Game](#) singleton wrapper ([getInstance\(\)](#), [run\(\)](#), game loop, window setup).
- Implement [SettingManager](#) singleton storing/loading game volumes, resolution, and control binds.
- Implement stack-based [StateManager](#) ([pushState\(\)](#), [popState\(\)](#), [changeState\(\)](#)).
- Implement abstract [GameState](#) class interface.
- Create concrete states: [MainMenuState](#) (Start, Load, Exit buttons), [GameState](#) (black window placeholder), [PauseState](#) (overlay), [GameOverState](#) (receives [EndingType](#) enum to display Ending A, B, or C).
- Planning on easy-to-use UI framework classes and color palette management.

Week 04

- Implement complete easy-to-use UI framework that supports [UI::FlexBox](#) for automatic position computation.
- Implement abstract [Character](#) base class (position, velocity, HP/maxHP, moveSpeed, baseDamage, attackRange, attackRate, sprite, [vector<unique_ptr<StatusEffect>> activeEffects](#), [takeDamage\(\)](#), [tickStatusEffects\(\)](#), [getEffectiveMoveSpeed\(\)](#)).
- Implement [Player](#) class inheriting from [Character](#). The player owns all three [PlayerForm](#) instances at all times ([unique_ptr](#) members) and holds a raw [PlayerForm*](#) pointing at whichever is active. Also track [switchCooldownTimer](#), [inMidChamber](#) flag (suspends the 4.0s cooldown when true), and the shared HP pool sized to the active form's [maxHP](#).
- Implement abstract [PlayerForm](#) Strategy base class with [attack\(\)](#), [useSpecial1\(\)](#), [useSpecial2\(\)](#), [onMomentumGainHit\(\)](#), [onMomentumGainDamageTaken\(\)](#), [addMomentum\(\)](#), [resetMomentum\(\)](#), and [setSpecial1Threshold\(\)](#) (called at runtime by [RunState](#) when Hollow Bell is collected).

- Code **WraithbladeForm** (Speed 7.0, MaxHP 100, Dmg 12, Range 1.5 melee, Rate 2/s; Momentum +6 per hit landed, +0.4 per Wraithblade-equivalent HP lost).
- Code **VoidcasterForm** (Speed 5.0, MaxHP 70, Dmg 22, Range 12 piercing, Rate 1/s; Momentum +8 per far-range hit, +4 per additional pierced enemy, +0.4 per HP lost).
- Code **IronshellForm** (Speed 2.5, MaxHP 160, Dmg 6, Range 1.0 cleave, Rate 1/s; Momentum +1.2 per Ironshell-equivalent HP lost, +3 per hit landed).
- Planning for seminar & overall plan updates.

25125061 - Châu Tấn Phát:

Week 01 & 02

- Create [implementation plan](#) as a detailed 7-week tasks lists, each week including tasks for 2 developers and tasks involving AI agents. Note that week count from week 3, so the first week of the plan is week 3 of the course.

Week 03

- Implement **AssetManager** singleton caching textures, fonts.
- Implement **SoundManager** singleton controlling background music streams and sound buffers.
- Implement **SaveLoadManager** singleton with **saveGame(RunState&)** and **loadGame() → RunState** interfaces; keep it decoupled from map data (that is **MapLoader**'s job).
- Define the **RunState** struct (fields: **currentLevel**, **currentChamber**, **map<EchoType, EchoOutcome> echoOutcomes**, **playerHP**, **activeForm**, per-form Momentum floats, **collectTimeReduction**, **special1MomentumThreshold**, **special2MomentumThreshold**, **foretellActive**, **foretellPhase1**). This struct is used by every subsequent system and must exist before Week 2 work begins.
- Integrate default UI font and basic game sound assets (menu navigation sounds, placeholder music).

Week 04

- Bind WASD/D-pad to player velocity.
- Implement **CollisionSolver** static utility class: **aabbIntersect()**, **circleIntersect()**, **resolveAABB()**, **lineIntersectsRect()**, and **lineIntersectsCircle()** (the last two are needed for Voidcaster's wall-piercing Special 1 in Week 5).
- Implement the switch key binds (1, 2, 3) to swap active **PlayerForm** strategies. On switch: reset the outgoing form's Momentum to 0, carry the incoming form's frozen Momentum, apply the HP conversion formula $\text{new_HP} = (\text{current_HP} / \text{old_max}) \times \text{new_max}$, and start the 4.0s **switchCooldownTimer**. Block switching if **switchCooldownTimer > 0** and **inMidChamber** is false. (Note: HP conversion removed, **inMidChamber** is determined by **Player::isSwitchCooldownEnabled**)
- Implement 4.0s switch cooldown timer that blocks swapping but does not inhibit movement or combat.

AI Usage Declaration

- **Claude**. Anthropic, accessed 19/06/2026. Purpose: Write a detailed game description with specific characters' stats from brief game description and rulesets.
- **Gemini**. Google, via Antigravity, accessed 11:23 AM 19/06/2026. Purpose: Design a detailed implementation plan given the game description and architecture.
- Other tasks done by AI agents using Gemini model, via Antigravity:

- Boilerplate CMake file configuration.
- Standard C++ singleton template class structure.
- Window initialization and rendering loop boilerplate.
- `EndingType`, `EchoType`, `EchoOutcome`, `FormType`, `ChamberType`, and `EnemyType` enum definitions (one shared header; all other classes include it).
- Implement reusable UI framework classes under folder `ui` inside namespace `UI` (`Component`, `Text`, `Button`, `Slider`, `Container`, `FlexBox<Axis>`, `HorizontalBox`, `VerticalBox`, `TextInput`) & `ColorPalette` struct, `ColorPaletteManager` singleton, and `ColorPaletteObserver` interface.
- Standard vector math utility functions (normalization, dot product, distance checks).
- Static class attributes and getters/setters for `Character` and `PlayerForm`.
- 2D AABB intersection checking code.

Issues

No issues encountered during this week.